Problem Statement

You are given a grid of dimensions N × M. Each cell contains one of the following values:

0 → Empty cell 1 → Fresh orange 2 → Rotten orange

A rotten orange can rot an adjacent fresh orange in 1 unit of time.

A rotten orange at position (i, j) can rot fresh oranges at:

(i-1, j) → Up (i+1, j) → Down (i, j-1) → Left (i, j+1) → Right

Find the minimum time required to rot all fresh oranges.

If it is impossible to rot all fresh oranges, print -1.

If there are no fresh oranges initially, the required time is 0.

Input Format

The first line contains an integer T, the number of test cases. For each test case: The first line contains two integers N and M, representing the number of rows and columns. The next N lines contain M space-separated integers representing the grid.

Constraints

1 ≤ T ≤ 100 1 ≤ N, M ≤ 500 grid[i][j] ∈ {0, 1, 2}

Output Format

Print a single integer representing the minimum time required to rot all fresh oranges.

Print -1 if some fresh oranges can never become rotten

Sample Input 0

```
3 3
2 1 1
1 1 0
0 1 1
```

Sample Output 0

```
4
```

Sample Input 1

```
3 3
2 1 1
0 1 1
1 0 1
```

Sample Output 1

```
-1
```

```python
# Enter your code here. Read input from STDIN.
from collections import deque
n, m = map(int, input().split())

grid = []
q = deque()
fresh = 0

for i in range(n):
    row = list(map(int, input().split()))
    grid.append(row)

    for j in range(m):
        if grid[i][j] == 2:
            q.append((i, j))
        elif grid[i][j] == 1:
            fresh += 1

time = 0

while q:
    size = len(q)

    for _ in range(size):
        i, j = q.popleft()

        for x, y in [(i-1,j), (i+1,j), (i,j-1)
```

Line: 41 Col: 1

Upload Code as File

Test against custom input

Run Code Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Input (stdin) Download

```
3 3
2 1 1
1 1 0
0 1 1
```

Your Output (stdout)

```
4
```

Expected Output Download

```
4
```